

Emily Tran

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Education

California State Polytechnic University, Pomona

Bachelor of Science in Computer Science, Minor in Data Science

GPA: 3.74 | Magna Cum Laude

Aug 2022 – May 2026

Pomona, CA

Projects

Gambler's Addict | *Unity, C#, Gameplay Systems, Adaptive AI*

- Designed and implemented adaptive gameplay, economy, and progression systems for a 2D bullet-hell roguelike in Unity
- Engineered modular AI behaviors and dynamic difficulty scaling systems
- Developed debugging and gameplay visualization tools to support balancing and playtesting iteration
- Built persistent progression systems including save-state management, cross-scene data persistence, and adaptive reward multipliers

Funk vs Fright | *Unity, C#, Animation Systems, Gameplay Systems*

- Led a 4-person interdisciplinary team in coordinating gameplay, art, and UI development for a 2.5D rhythm-combat game in Unity
- Implemented timing-sensitive gameplay systems, animation state management, and gameplay feedback loops
- Created UI assets, sprite animations, and visual feedback systems to improve gameplay readability
- Awarded 1st Place overall in a competitive game jam

Calendar Meet | *React, Node.js, PostgreSQL, AWS, Github Actions*

- Collaborated on development of a full-stack scheduling platform
- Implemented cloud deployment infrastructure on AWS and automated CI/CD pipelines using GitHub Actions
- Maintained application deployments, database integrations, and production updates

Experience

Virtual Reality Lab — Cal Poly Pomona

Mar 2025 – Jun 2025

Unity Developer

Pomona, CA

- Collaborated within a 4-person multidisciplinary team to develop a VR zombie survival FPS built in Unity using XR Interaction Toolkit
- Developed VR interactive systems, gameplay features, enemy AI behaviors, and spatial audio functionality
- Tested, debugged, and refined features through iterative development and playtesting

Research Experience for Undergraduates (REU) — Texas A&M University

May 2024 – Aug 2024

Applied Computational Robotics Researcher

College Station, TX

- Developed Pygame simulations to evaluate autonomous rendezvous algorithms
- Implemented and analyzed A* and D* Lite pathfinding algorithms
- Conducted comparative testing and presented technical findings in a collaborative research environment

Technical Skills

Languages: C#, Python, Java, JavaScript

Frameworks & Technologies: Unity, React, Node.js, PostgreSQL, AWS

Game Development: Gameplay Systems, Game AI, XR Interaction Toolkit

Developer Tools: Git, GitHub, GitHub Actions, VS Code, Aseprite